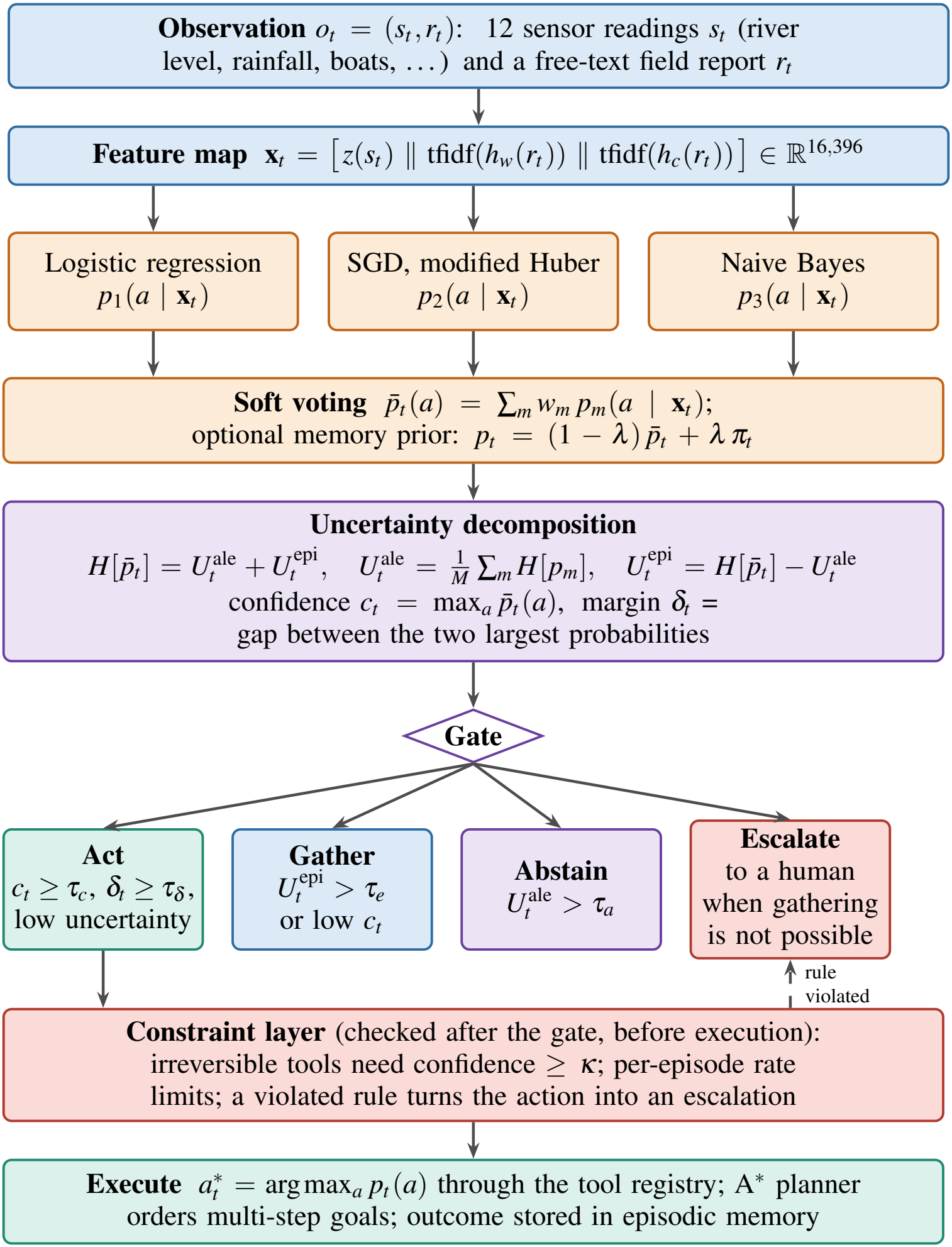

Observation $o_t = (s_t, r_t)$: 12 sensor readings s_t (river level, rainfall, boats, ...) and a free-text field report r_t
Feature map $\mathbf{x}_t = [z(s_t) \parallel \text{tfidf}(h_w(r_t)) \parallel \text{tfidf}(h_c(r_t))] \in \mathbb{R}^{16,396}$
Logistic regression $p_1(a \mid \mathbf{x}_t)$
SGD, modified Huber $p_2(a \mid \mathbf{x}_t)$
Naive Bayes $p_3(a \mid \mathbf{x}_t)$
Soft voting $\bar{p}_t(a) = \sum_m w_m p_m(a \mid \mathbf{x}_t)$; optional memory prior: $p_t = (1-\lambda)\bar{p}_t + \lambda \pi_t$
Uncertainty decomposition
$H[\bar{p}_t] = U_t^{\text{ale}} + U_t^{\text{epi}}$, $U_t^{\text{ale}} = \frac{1}{M}\sum_m H[p_m]$, $U_t^{\text{epi}} = H[\bar{p}_t] - U_t^{\text{ale}}$
confidence $c_t = \max_a \bar{p}_t(a)$, margin δ_t = gap between the two largest probabilities
Gate
Act
$c_t \geq \tau_c$, $\delta_t \geq \tau_\delta$, low uncertainty
Gather
$U_t^{\text{epi}} > \tau_e$ or low c_t
Abstain
$U_t^{\text{ale}} > \tau_a$
Escalate
to a human when gathering is not possible
rule violated
Constraint layer (checked after the gate, before execution): irreversible tools need confidence $\geq \kappa$; per-episode rate limits; a violated rule turns the action into an escalation
Execute $a_t^* = \arg\max_a p_t(a)$ through the tool registry; A* planner orders multi-step goals; outcome stored in episodic memory